CS & IT ENGINEERING

Operating System

Process Management

Lecture -1

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Recap of Previous Lecture

Topic Operating System Definition

Topic Types of Operating System

Topics to be Covered

Topic Dual Mode of Operation

Topic Process

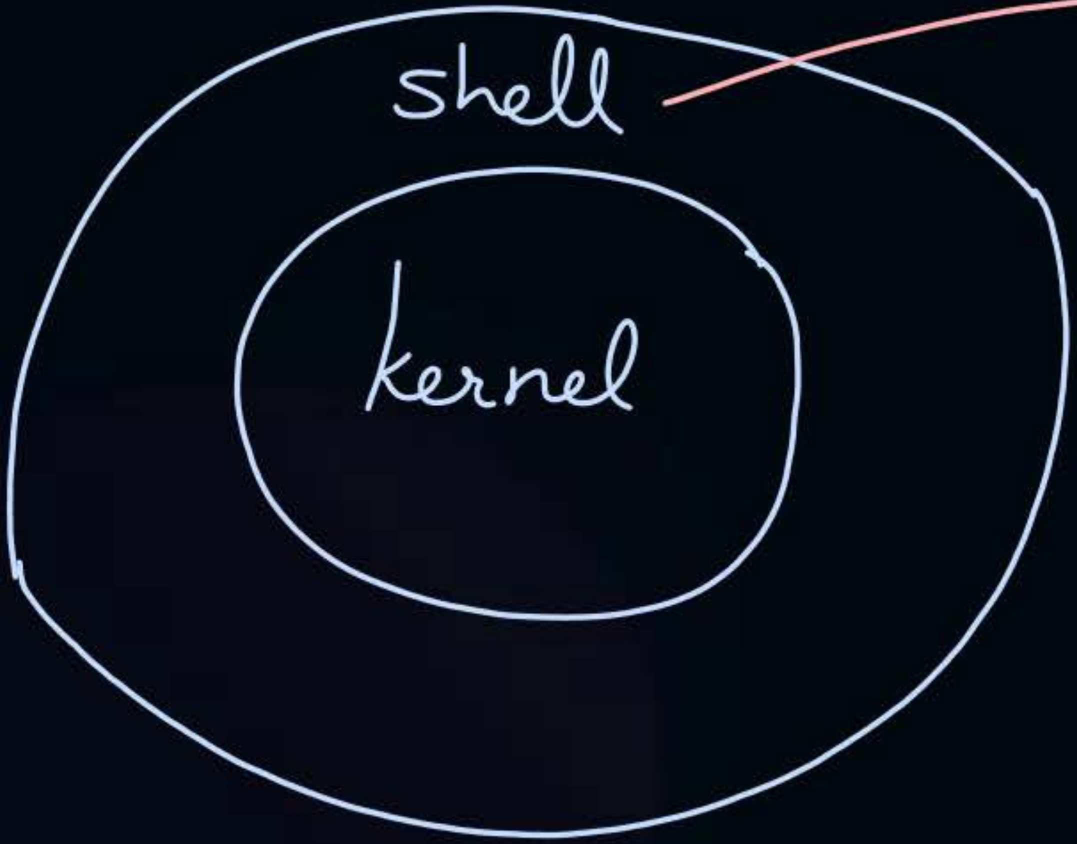
Topic Process Representation

Topic Process Control Block

Topic : System Call

A system call is a way for programs to interact with the operating system

Topic : Parts of OS



→ GUI (Graphical User Interface)

→ CLI (Command Line Interpreter)

Topic : Dual Mode of Operation

2 Modes:

User Mode (mode bit = 1)

Kernel/System/Supervisor/Privileged Mode (mode bit = 0)

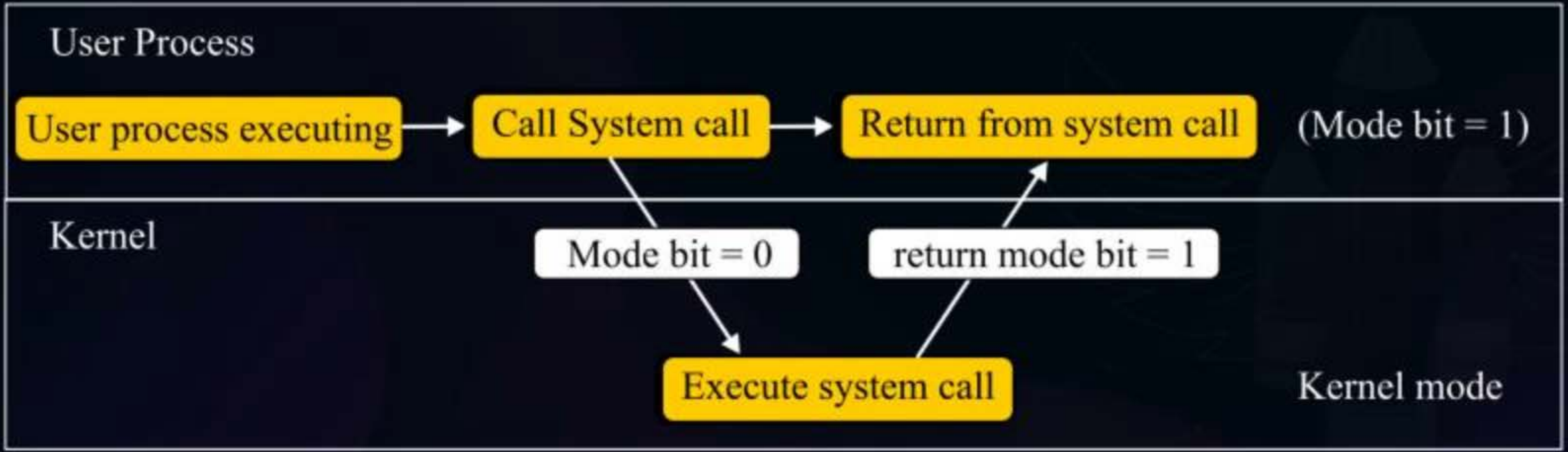
Topic : Dual Mode of Operation

To provide protection

2 Modes:

User Mode (mode bit = 1)

Kernel/System/Supervisor/Privileged Mode (mode bit = 0)



Topic : Process

A running program called as process.

Process = Program + Runtime activity

Topic : Process

- **Process:**
  - Program under execution
  - An instance of a program
  - Schedulable/Dispatchable unit (CPU)
  - Unit of execution (CPU)
  - Locus of control (OS)

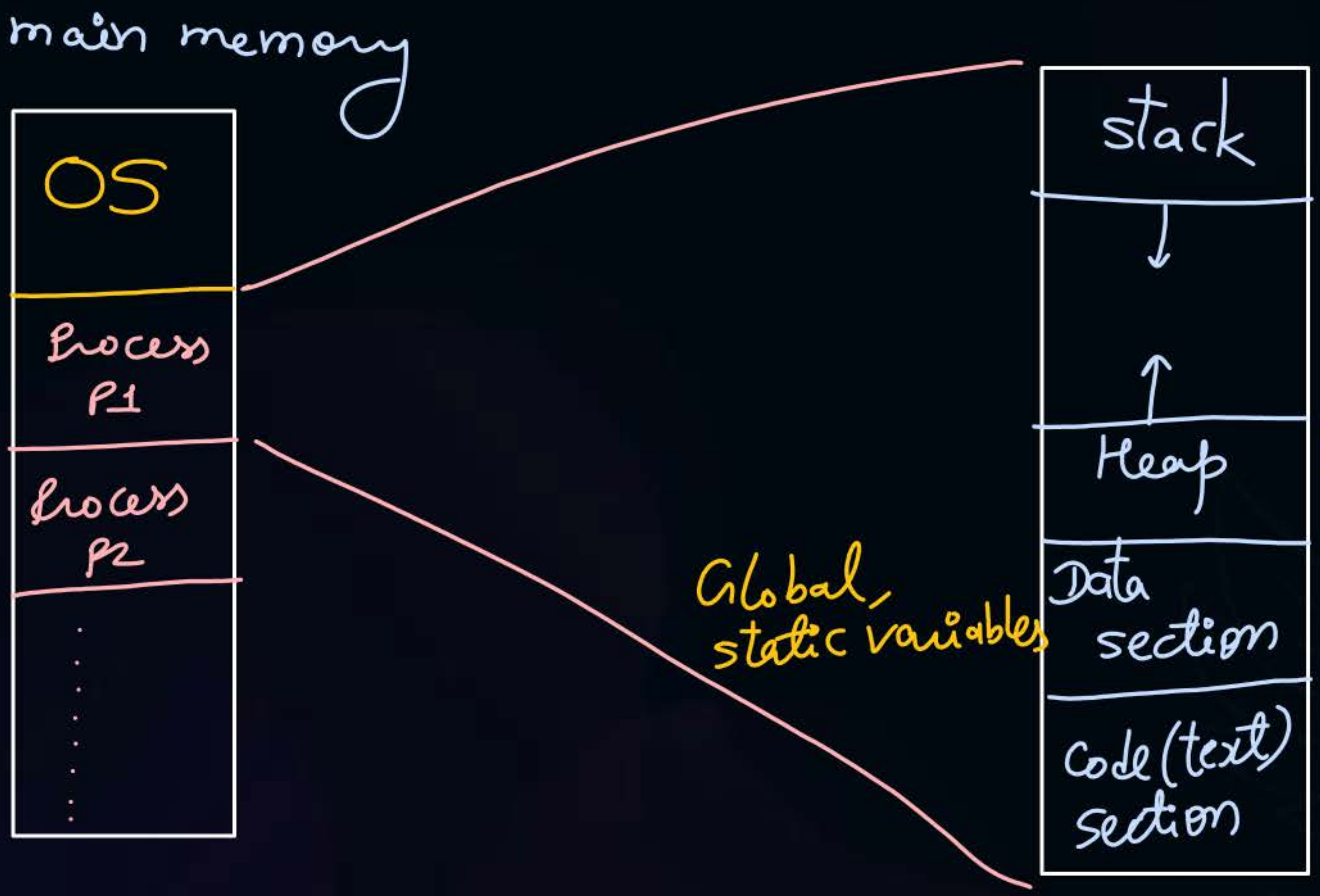
Topic : Process

Process

- Definition
- Representation/ Implementation
- Operations
- Attributes

Topic : Representation of a Process

→ how each process stored in m.m.



Activation Record (return address, local variables, function parameters)

Dynamic memory allocation

Program instruction

malloc() or calloc() do not need system call.

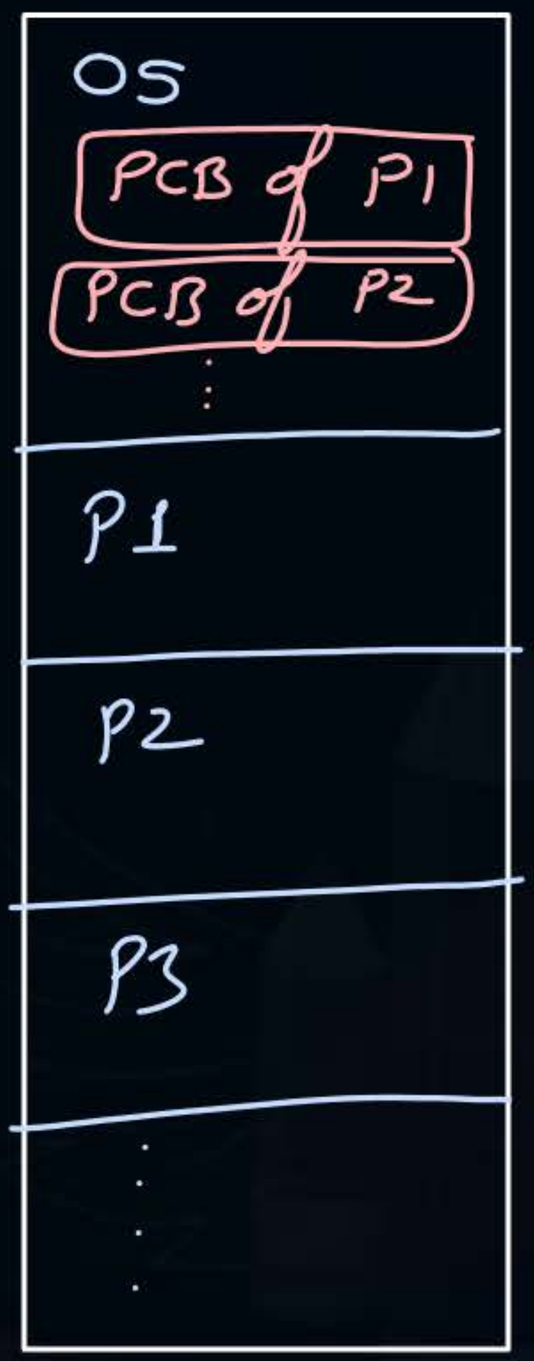
Topic : Operations on a Process

- Create (Resource Allocation)
- Schedule, Run
- Wait/Block
- Suspend, Resume
- Terminate (Resource Deallocation)

Topic : Attributes of a Process

- PID (Process id)
- PC (Program counter value)
- GPR (General Purpose Registers)
- List of Devices
- Type
- Size
- Memory Limits
- Priority
- State
- List of Files

Main memory



Topic : PCB

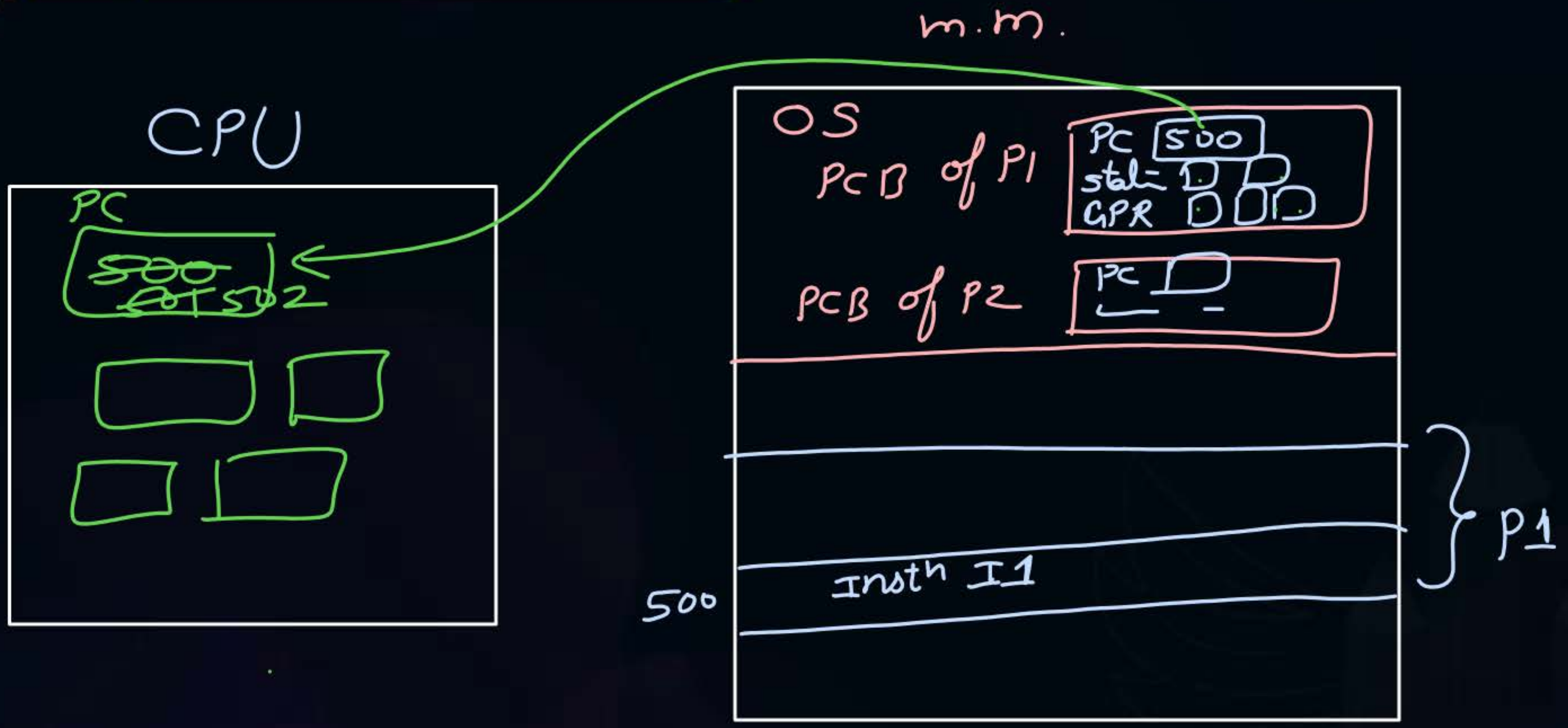
Also known as processor descriptor

PCB stores attributes of a process.

Topic : Context

The content of PCB of a process are collectively know as 'Context' of that process

Topic : Context Switch



Context switch :-

→ 1. context save of current running process from CPU to it's PCB

→ 2. Context load of next process from it's PCB to CPU

Dispatcher :- performs context switch

[NAT]

#Q. While running, a process can access its PCB from main memory?

NO

[MCQ]

#Q. A process in the context of computing is:

A. A set of instructions to be executed on a computer ⇒ Program

B. A program in execution

C. A piece of hardware that executes a set of instructions ⇒ CPU

D. The main procedure of a program ⇒ main function

2 mins Summary

Topic Dual Mode of Operation

Topic Process

Topic Process Representation

Topic Process Control Block

Happy Learning

THANK - YOU